## DTC P0101:00 [SKYACTIV-D 1.5]

id0102q2700800

| DTC P0101:00 | MAF sensor circuit range/performance problem |
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| **DETECTION CONDITION** | • When the following conditions are met, the intake air amount is out of the specification (fluctuates with engine speed) for a continuous **5.2 s**.<br>**MONITORING CONDITIONS**<br>— Battery voltage: **8—20 V**<br>— **60 s** have elapsed after enigine was started.<br>— Engine speed: **1,000—3,500 rpm**<br>— Fuel injection amount: **0 $mm^3$/st**<br>— Charge air temperature: **below 65 °C {149 °F}**<br>— Desired HP-EGR valve position: **below 1 %**<br>— Desired intake shutter valve position: **80 °**<br>— Intake manifold pressure: **below 200 kPa {2.03 $kgf/cm^2$, 29.0 psi}**<br>— Duty value of the suction control valve: **above 32 %**<br>— IAT sensor No.2: Normal<br>— BARO sensor: Normal<br>— The following DTCs are not detected:<br>• HP-EGR control valve position sensor: P0405:00, P0406:00<br>• HP-EGR control valve DC motor control system: P0404:00, P0488:00<br>• Intake shutter valve control: P2101:00ÅAP2118:00ÅAP1589:00<br>• MAF sensor: P0102:00, P0103:00<br>• IAT sensor No.2: P007C:00, P007D:00<br>• MAP sensor: P0107:00, P0108:00<br>• Intake sutter valve position sensor: P0122:00, P0123:00<br>• BARO sensor: P2228:00, P2229:00<br>**Diagnostic support note**<br>• This is a continuous monitor (CCM).<br>• The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle.<br>• FREEZE FRAME DATA/Snapshot data is available.<br>• DTC is stored in the PCM memory. |
| **FAIL-SAFE FUNCTION** | • PCM restricts engine torque.<br>• Inhibits the EGR control.<br>• Inhibits the auto DPF regeneration control.<br>• Inhibits the compulsory DPF regeneration control.<br>• Inhibits engine-stop by operating the i-stop function. |
| **POSSIBLE CAUSE** | • MAF sensor/IAT sensor No.1 connector or terminals malfunction<br>• MAF sensor/IAT sensor No.1 loose<br>• PCM connector or terminals malfunction<br>• Air suction or restriction in intake-air system<br>• Exhaust gas leakage from exhaust system or clogging (high pressure EGR)<br>• Exhaust gas leakage from exhaust system or clogging (low pressure EGR)<br>• MAF sensor malfunction<br>• Turbocharger malfunction (turbine wheel and/or compressor wheel damaged, stuck)<br>• PCM malfunction |
| **SYSTEM WIRING DIAGRAM** | Not applicable |

### Diagnostic Procedure

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION**<br>**Note**<br>• Recording can be facilitated using the screen capture of the PC function.<br>• Record the freeze frame data/snap shot data. | — | Go to the next step. |
| 2 | **VERIFY RELATED REPAIR INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 3 | **VERIFY RELATED PENDING CODE AND/OR DTC**<br>• Switch the ignition off, then ON (engine off).<br>• Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Are any other PENDING CODEs and/or DTCs present? | Yes | Go to the applicable PENDING CODE or DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 4 | **INSPECT MAF SENSOR/IAT SENSOR NO.1 CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the MAF sensor/IAT sensor No.1 connector.<br>• Inspect for poor connection (such as damaged/pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 13. |
| | | No | Go to the next step. |
| 5 | **INSPECT INSTALLATION OF MAF SENSOR/IAT SENSOR NO.1**<br>• Inspect installation of MAF sensor/IAT sensor No.1.<br>(See MASS AIR FLOW (MAF) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>• Is the MAF sensor/IAT sensor No.1 installed securely? | Yes | Go to the next step. |
| | | No | Retighten the MAF sensor/IAT sensor No.1, then go to Step 13.<br>(See MASS AIR FLOW (MAF) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| 6 | **INSPECT PCM CONNECTOR CONDITION**<br>• Disconnect the PCM connector.<br>• Inspect for poor connection (such as damaged/pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 13. |
| | | No | Go to the next step. |
| 7 | **INSPECT INTAKE AIR SYSTEM FOR EXCESSIVE AIR SUCTION**<br>• Visually inspect for loose, cracked or damaged hoses on intake air system.<br>**Note**<br>• Engine speed may change when rust penetrating agent is sprayed on the air suction area.<br>• Is there any malfunction? | Yes | Repair or replace the malfunctioning part according to the inspection results, then go to Step 13. |
| | | No | Go to the next step. |
| 8 | **INSPECT FOR RESTRICTION IN INTAKE AIR SYSTEM**<br>• Verify if there is restriction into the intake air system (such as between MAF sensor and intake manifold).<br>• Is there any malfunction? | Yes | Repair or replace the malfunctioning part according to the inspection results, then go to Step 13. |
| | | No | Go to the next step. |
| 9 | **INSPECT EXHAUST SYSTEM (HIGH PRESSURE EGR) RELATED PIPE**<br>• Visually inspect the exhaust system (high pressure EGR) related pipe for restriction and damaged.<br>(See EMISSION SYSTEM LOCATION INDEX [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Repair or replace the malfunctioning part according to the inspection results, then go to Step 13. |
| | | No | Go to the next step. |
| 10 | **INSPECT EXHAUST SYSTEM (LOW PRESSURE EGR) RELATED PIPE**<br>• Visually inspect the exhaust system (low pressure EGR) related pipe for restriction and damaged.<br>(See EMISSION SYSTEM LOCATION INDEX [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Repair or replace the malfunctioning part according to the inspection results, then go to Step 13. |
| | | No | Go to the next step. |
| 11 | **INSPECT MAF SENSOR**<br>• Reconnect all disconnected connectors.<br>• Inspect the MAF sensor.<br>(See MASS AIR FLOW (MAF) SENSOR INSPECTION [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Replace the MAF sensor/IAT sensor No.1, then go to Step 13.<br>(See MASS AIR FLOW (MAF) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 12 | **INSPECT TURBOCHARGER**<br>• Inspect the turbocharger.<br>(See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].)<br>• Is the turbocharger normal? | Yes | Go to the next step. |
| | | No | Replace the turbocharger, then go to Step 13.<br>(See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |

| STEP | INSPECTION | | ACTION |
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| 13 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].)<br>• Start the engine.<br>• Repeat deceleration 10 times using engine braking after accelerating the vehicle speed to **60 km/h {37 mph}**.<br>• Perform the Pending Trouble Code Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the PENDING CODE for this DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 14 | **VERIFY AFTER REPAIR PROCEDURE**<br>• Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | DTC troubleshooting completed. |